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Maternal steroids during pregnancy and their associations with ambient air pollution and temperature during preconception and early gestational periods

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Abstract

Hormones play critical roles in facilitating pregnancy progression and the onset of parturition. Several classes of environmental contaminants, including fine particulate matter (PM_{2.5}) and ambient temperature, have been shown to alter hormone biosynthesis or activity. However, epidemiologic research has not considered PM_{2.5} in relation to a broader range of steroid hormones, particularly in pregnant women. Using metabolomics data collected within 20–40 weeks of gestation in an ethnically diverse pregnancy cohort study, we identified 42 steroid hormones that we grouped into five classes (pregnenolone, androgens, estrogens, progestin, and corticosteroids) based on their biosynthesis type. We found that exposure to PM_{2.5} during the pre-conception and early prenatal periods was associated with higher maternal androgen concentrations in late pregnancy. We also detected a positive association between early pregnancy

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CRediT authorship contribution statement

Elena Colicino: Conceptualization, Methodology, Investigation, Writing – original draft. **Whitney Cowell:** Methodology, Writing – review & editing. **Nicolo Foppa Pedretti:** Formal analysis, Visualization. **Anu Joshi:** Formal analysis, Visualization. **Oulhote Youssef:** Writing – review & editing. **Allan C. Just:** . **Itai Kloog:** . **Lauren Petrick:** Writing – review & editing. **Megan Niedzwiecki:** Writing – review & editing. **Robert O. Wright:** . **Rosalind J. Wright:** Conceptualization, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envint.2022.107320>.

PM_{2.5} exposure and maternal pregnenolone levels and a marginal positive association between early pregnancy PM_{2.5} exposure and progesterone levels. When considering each hormone metabolite individually, we found positive associations between early pregnancy PM_{2.5} exposure and five steroids, two of which survived multiple comparison testing: 11beta-hydroxyandrosterone glucuronide (a pregnenolone steroid) and androstenedione glucuronide (a progesterone steroid). None of the steroid classes were statistically significant associated with ambient temperature. In sex-stratified analyses, we did not detect any sex differences in our associations. This is the first study showing that exposure to fine particulate matter during the pre-conception and early prenatal periods can lead to altered steroid adaptation during the state of pregnancy, which has been shown to have potential consequences on maternal and child health.

Keywords

Maternal steroids; Air pollution; Temperature; Preconception exposure; Prenatal exposure

1. Introduction

Maternal steroid hormones play a critical role in the development of the fetus and progression of parturition. For example, steroid hormones drive modifications occurring in the uterine microenvironment throughout gestation and at term, leading to an increase in uterine size and successful development of the placenta (Chang and Zhang, 2008). Maternal steroids also enable the uterus and placenta to improve vascularization, thereby enhancing the transfer of nutrients (Chang and Zhang, 2008), and facilitating the onset and progression of parturition (Dehghan et al., 2013). Generally, steroidogenesis increases throughout gestation in order to support evolving physiological demands that improve the likelihood of a successful pregnancy (Solano and Arck, 2020). Sub-optimal endocrine adaptation to the state of pregnancy can impact maternal health and perinatal outcomes, and have longer-term consequences for child health. For example, insufficient or excessive hormonal shifts during pregnancy can contribute to impaired fetal growth, preterm birth, miscarriage, preeclampsia, and gestational diabetes mellitus, with longer-term concerns including thyroid dysfunction and adverse neurodevelopmental outcomes (Haddow et al., 1999; Henrichs et al., 2010; Korevaar et al., 2016; Medici et al., 2013; Millar et al., 1994; Noyola-Martínez et al., 2019; Oken et al., 2009; Phoojaroenchanachai et al., 2001; Pop et al., 2003; Pop et al., 1999).

Several classes of environmental contaminants are defined as endocrine disrupting chemicals (EDC), including dioxins, polychlorinated biphenyls, pesticides, and heavy metals as they interfere directly or indirectly with hormone biosynthesis, metabolism or activity (Freire et al., 2011; Iijima et al., 2007; Wang et al., 2005). Notably, recent research has linked particulate air pollution with disruption of maternal thyroid hormones in later pregnancy or at birth (Ghassabian et al., 2019; Howe et al., 2018; Zhao et al., 2019). Fine particulate matter (PM_{2.5}), which has an aerodynamic diameter <2.5 µm can penetrate deep lung alveoli and even enter the blood stream. PM_{2.5} has obesogenic properties and may be linked to disruption of metabolic processes under hormonal control (Heindel and Schug, 2014). Despite these early findings and the ubiquity of PM_{2.5} exposure, epidemiologic research to

date has not considered PM_{2.5} in relation to a broader range of steroid hormones, i.e. up to 42 steroids, particularly in pregnant women.

Untargeted metabolomics assays allow for the concurrent detection of thousands of small molecules in human biofluids, including steroid hormones (Niedzwiecki et al., 2020; Niedzwiecki et al., 2019). Herein we leveraged a longitudinal ethnically diverse urban pregnancy cohort (n = 361) based in the Northeastern United States to examine exposure to PM_{2.5} during the pre-conception and early gestation periods in relation to maternal serum hormones during later pregnancy assessed using an untargeted metabolomics assay. We considered hormones both grouped into five classes (androgenic steroids, estrogenic steroids, pregnenolone steroids, progestin steroids, and corticosteroids) and individually (total number of hormones = 42) and used robust mixed effect model regressions for the steroid classes, while for individual steroids we used robust regressions. While our primary interest was in associations between PM and maternal hormone concentrations during late pregnancy, we additionally examined associations with ambient temperature. Indeed, temperature co-varies with PM exposure and can also modulate both hormone release (Bridge et al., 2003) and broader endocrine control (Collins, 1978), but it has not previously been studied in relation to maternal hormone levels during pregnancy. Finally, we considered the modifying role of sex in these relationships given that fetal sex can influence the maternal system during pregnancy (Al-Qaraghoul and Fang, 2017), multiple physiologic systems show sexually-dimorphic responding to environmental exposures (Al-Qaraghoul and Fang, 2017; Cowell and Wright, 2017) and a number of pregnancy-related and perinatal health outcomes vary by sex (Al-Qaraghoul and Fang, 2017),.

2. Methods

2.1. Study sample

The PRogramming of Intergenerational Stress Mechanisms (PRISM) study is a prospective pregnancy cohort designed to investigate the role of early life environmental exposures, including ambient air pollution and temperature levels, in relation to pregnancy outcomes and child development. We enrolled women, who were within the 11–40 weeks of gestation, from prenatal clinics in Boston and New York City beginning in 2011. All eligible participants were English or Spanish-speakers, over 18 years of age at enrollment, and had a singleton pregnancy. Due to the association of HIV status and alcohol consumption with birth outcomes, we excluded mother–child pairs with HIV + status and mothers that self-reported drinking ≥ 7 alcoholic drinks per week before pregnancy or any alcohol after pregnancy recognition. The current analyses include 361 participants with metabolomics data, air pollution and temperature levels, fetal sex, and complete information on a core set of covariates (Figure S1, Table S1).

2.2. Ethics

All study protocols were approved by the human studies' committees at the Brigham and Women's Hospital (BWH) in Boston and the Icahn School of Medicine at Mount Sinai (ISMMMS) in New York City. All participants provided written informed consent in their primary language of preference.

2.3. Maternal steroid identification: Untargeted metabolomics phenotyping during pregnancy

Maternal blood was collected once by venipuncture (mean $\pm$ SD: 29.5 $\pm$ 4.90 weeks, Table S2) and serum aliquots were stored at -80°C until assayed. We measured maternal steroids during pregnancy as part of an untargeted metabolomics profiling assay. As previously described (Colicino et al., 2021), untargeted metabolomics analysis was conducted on 100 μl of serum, at Metabolon, Inc (Durham, NC, USA) using an ultra-performance liquid chromatography (UPLC) and a Thermo Scientific Q-Exactive high resolution/accurate mass spectrometer interfaced with a heated electrospray ionization (HESI-II) source and Orbitrap mass analyzer. Raw data were extracted, peak-identified and quality control processed using Metabolon's hardware and software. Batch adjustment to correct variation resulting from instrument inter-day tuning differences was performed for each compound in run-day blocks by dividing by the median of the values for the experimental samples for each instrument run day, then multiplying these values by the original median. In total, 1,100 biochemicals were measured and annotated by comparison with purified standards of the Metabolon Library, when available. Of these biochemicals, we found 42 steroid hormones, classified as androgenic steroids, estrogenic steroids, corticosteroids, progestin steroids, and pregnenolone steroids. All abundance values were presented as normalized peak areas to facilitate both linear and non-linear analyses and to harmonize all variables on a common scale. After the pre-processing and cleaning steps, we did not identify any systematic bias in our metabolomics data using principal component analysis (Figure S2).

2.4. Fine air particles and temperature levels before conception and during early pregnancy.

Predicted daily ambient pollutant concentrations and temperature during the preconception and early gestation periods (first 20 weeks) were derived for each participant based on their validated geocoded residential address at enrollment. All addresses were then updated to account for any moves at all follow-up visits. We computed exposures of participants with multiple addresses as the average exposure estimates between multiple locations. Estimated residence-specific daily $\text{PM}_{2.5}$ and temperature exposures three months before conception and over the first 20 weeks of pregnancy were considered in analyses. As previously detailed (Carrión et al., 2020; Just et al., 2020), we regressed daily surface $\text{PM}_{2.5}$ and temperature measurements obtained by U.S. Environmental Protection Agency Air Quality System and Interagency Monitoring of Protected Visual Environments Network with satellite-derive aerosol optical depth measurements (1 km^2 spatial resolution). To determine residence-specific daily $\text{PM}_{2.5}$ and temperature levels, we included meteorological variables (seasonality) and land-use terms (inverse distance weighted PM monitor surface, percentage of developed areas, vegetation index) in machine learning algorithms, which helped to minimize the prediction error and accurately estimate the daily exposures (Carrión et al., 2020; Just et al., 2020). We calibrated our daily prediction models and validated the estimates with a robust out of sample 10-fold cross-validation ($R^2 = 0.88$ for air pollution and $R^2 = 0.98$ for temperature) (Carrión et al., 2020; Just et al., 2020). For these analyses, we averaged daily $\text{PM}_{2.5}$ and daily temperature exposures over the pre-conception period of three months and over the first 20 weeks of gestation, due to their impact on the maternal endocrine system at large. (Braun et al., 2017; Dechanet et al., 2011; Howe et al., 2018;

Siegel et al., 1995; Toivonen et al., 2017) Both pre-conceptional and prenatal periods are increasingly recognized as a highly sensitive window for environmental perturbation, via both maternal and paternal exposures (Braun et al., 2017; Dechanet et al., 2011; Siegel et al., 1995; Toivonen et al., 2017). Pre-conception exposures were linked to alterations of hormonal balance and epigenetic programming of reproductive tissues able to exert hormones, while early prenatal exposures have been associated with adverse maternal and fetal endocrine outcomes (Luderer et al., 2019; Mustieles et al., 2020; Nevoral et al., 2018).

2.5. Covariates

Participant age (continuous in years), self-reported race/ethnicity (White, non-Hispanic/other vs. Black/Black-Hispanic vs. Hispanic, non-Black), and education level (<high school vs. high school or more) were determined by questionnaires administered during a standardized in-person interview during pregnancy. Gestational week of serum collection was calculated based on the difference between the best obstetric estimate of gestational age at delivery and date of collection. Fetal sex was recorded based on review of electronic medical records.

2.6. Statistical analysis

Throughout gestation, fluctuations of steroid levels contribute to a coordinated sequence of events that must occur in order to establish and successfully maintain a healthy pregnancy. Indeed, some steroids have been shown to increase during pregnancy, such as estrogens, while others have a decreasing trend (Vannuccini et al., 2016). Across the last 20 weeks of gestation, individual steroids showed differences in concentrations due to gestational weeks during which the samples were collected (Figures S3). Prior literature supported these hormonal trends throughout gestation (Vannuccini et al., 2016). To remove the time-dependent pattern of hormonal change across gestation, which we considered a nuisance factor with regard to our primary research aim, and to have a common scale across all steroids, we standardized all steroids based on their weekly population average and standard deviation levels (Figures S4.A, Table S3). We considered these standardized steroid levels as the main outcome in our analyses. We assessed Spearman rank correlations between all standardized steroid levels. Due to the strong correlation among steroids and to reduce any effect of outliers of the steroids—defined as three standard deviations (SDs) far from the mean—we treated steroids in each class (androgenic, estrogenic, corticosteroids, progestin, and pregnenolone steroids) as ‘repeated’ steroid measures in robust mixed effect regression models. These models included a random intercept for individual to account for the different (i.e., ‘repeated’) steroid measures within individuals and a separate model was run for each class of steroid. Using this approach, we identified the association between $PM_{2.5}$ and temperature exposure, considered in the same model, with steroid levels in each class. In all models, we considered both exposures ($PM_{2.5}$ and temperature) as the average during the three months before conception and during the first 20 weeks of gestation. All models were adjusted for maternal age, ethnicity, and education. We did not adjust for smoking status because only two mothers smoked during pregnancy. We then performed a steroid-by-steroid analysis by fitting robust regression models between each steroid and both $PM_{2.5}$ and temperature levels. We inspected and presented results using volcano plots of the estimated coefficients and p-values. We corrected all results for multiple testings using a 1% Bonferroni-threshold for correcting for multiple steroid classes and a 5%

FDR-adjusted p-value for correcting for multiple individual steroids. Finally, we evaluated potential sex-specific effects by examining associations stratified by sex and by including the cross-product term between sex and our ambient exposures.

2.6.1. Sensitivity analyses.—We also evaluated the differences in the characteristics of the participants included in the main analysis compared to the cohort remainder (Table S1). To attenuate any potential selection bias, in our analysis we included weights constructed with the inverse probability weighting technique. This approach identifies the probability of being included in the study linking the covariates with a binary outcome indicating the inclusion or not in the study. We then winsorised metabolites levels above 95th and below 5th percentiles and we fitted linear mixed effect regressions for the steroid classes and robust regression models for individual steroids.

We finally evaluated the main associations including maternal pre-pregnancy body mass index (BMI) in the set of covariates due to its relationship with steroid levels.

3. Results

3.1. Study sample

Participants had an average age (SD) of 31 (5.6) years and were ethnically diverse (White, non-Hispanic/other 37%; Black/Black-Hispanic 30%; Hispanic, non-Black 33%). Out of the 361 mother–child pairs involved in the study, 186 (51.5%) women carried a male fetus. The majority (79%) of participants had a high school education or more (Table 1). The average PM_{2.5} exposure was 9.20 µg/m³ (SD: 1.82; interquartile range IQR: 1.91) during the three months before conception and 8.89 µg/m³ (SD: 1.30; IQR: 1.74) during the first 20 week of gestation. The average outdoor temperature levels were 11.85 °C (SD: 8.11; IQR: 14.73) during preconception and 10.43 °C (SD: 6.54; IQR: 12.01) during the first 20 weeks of gestation (Table 1). We present results in terms of a 2 µg/m³ change in PM_{2.5} and 10 °C change in temperature, which are approximately the exposure IQRs.

3.2. Steroid levels

We selected 42 steroids among 1,110 metabolites measured in serum collected during pregnancy, which we classified into five groups based on their biosynthesis type. Across the last 20 weeks of gestation, steroid levels showed time-dependent patterns (Figure S3). These results are consistent with prior literature showing linear, non-linear and both decreasing and increasing hormonal trends throughout gestation. As described in the methods, we removed the time-dependent pattern and created a common scale across steroids using a standardization procedure (Figures S4.A, Table S3). After this step, steroid levels had strong correlations within classes, with low- or anti-correlations between classes (Figure S4.B).

3.3. Associations of ambient air pollution and temperature with steroid classes

A 2 µg/m³ increase in PM_{2.5} exposure during early gestation was associated with increased levels of pregnenolone (Est.: 0.17; 95% Confidence Interval (CI): 0.05; 0.29, p-value: 0.007; Fig. 1) and androgen (Est.: 0.15; 95 %CI: 0.06; 0.25, p-value: 0.001; Fig. 1) steroids. A marginal association was also identified between early pregnancy PM_{2.5} exposure and

progesterone levels (Est.: 0.11; 95% CI: -0.01; 0.23; p-value: 0.07; Fig. 1). A 2 $\mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$ exposure in the three months before conception was also associated with an increase in androgen steroid levels (Est.: 0.08; 95% CI: 0.01; 0.14, p-value: 0.025; Fig. 1). Associations during early gestation, but not preconception, remained statistically significant after adjusting for multiple comparisons (Bonferroni correction threshold = 0.01). Androgens also showed a negative borderline association with temperature before conception (Est.: -0.08; 95% CI: -0.15; 0.00, p-value: 0.04; Fig. 1); however, this finding did not survive the multiple testing correction threshold.

3.4. Associations of ambient air pollution and temperature with individual steroids

When considering each steroid individually, rather than grouped as a class, we found that a 2 $\mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$ exposure during early pregnancy was positively associated with the concentrations of five steroids: two pregnenolone steroids (C62311: 11 β -hydroxyandrosterone glucuronide; C38170: pregnenolone sulfate), an androgen steroid (C37203: androstenediol disulfate), and two progesterone steroids (C4394: androstenediol glucuronide; C46172: 17 α -hydroxyandrosterone glucuronide) (Fig. 2.A; Table S4). Findings with C62311 (Est.: 0.29, FDR-adjusted p-value: 0.03) and C4394 (Est.: 0.29; FDR-adjusted p-value: 0.03) remained statistically significant at 5% FDR (Fig. 2.A; Table S4).

With regard to temperature, we found a negative association with preconception temperature and the previously identified androgen C37203 (Est.: -0.132; p-value: 0.044) and a positive association with early pregnancy temperature and the androgen C37187 (5 α -androstane-3 β , 17 α -diol disulfate) (Est.: 0.265; p-value: 0.001); however, the associations with temperature did not survive correction for multiple testing (Fig. 2.B; Table S4).

3.5. Sex-stratified analyses for steroid classes and individual steroids

3.5.1. Steroid classes—The positive associations between early pregnancy $\text{PM}_{2.5}$ exposure and (i) pregnenolone steroids (male: Est.: 0.14; 95 %CI: 0.02;0.27; p-value: 0.02; female: 0.19; 95% CI: 0.06,0.32; p-value: 0.004; p-for-interaction: 0.45; Fig. 3) and (ii) androgen steroids (male: Est.: 0.15; 95% CI: 0.05, 0.25; p-value: 0.004; female: Est.: 0.15; 95 %CI: 0.06, 0.25; p-value: 0.002; p-for-interaction: 0.40; Fig. 3) were similar in magnitude and direction across fetal sex. Likewise, the positive association between preconception $\text{PM}_{2.5}$ exposure and androgen steroids was also similar between male and female pregnancies (male: Est.: 0.08; 95% CI: 0.01, 0.15; p-value: 0.03; female: 0.08; 95% CI: 0.01, 0.15; p-value: 0.02; p-for-interaction: 0.65; Fig. 3). The associations between early pregnancy $\text{PM}_{2.5}$ and androgens remained statistically significant after adjusting for multiple comparisons for both fetal sexes; however, the association with pregnenolone remained statistically significant only for mothers carrying a female. The association between preconception $\text{PM}_{2.5}$ exposure and androgens did not survive multiple comparisons correction for either sex. Notably, we also identified an additional positive association between exposure to $\text{PM}_{2.5}$ during the early prenatal period and progesterone levels in mothers pregnant with a male (male: Est.: 0.15; 95% CI: 0.03, 0.28; p-value: 0.02; Fig. 3), which could have driven the marginal association observed when considering male and female

pregnancies together. The negative association between preconception temperature and androgen levels remained statistically significant among female pregnancies only (Est.: -0.09; 95% CI: -0.17, -0.01; p-value: 0.02 Fig. 3); however, this finding still did not survive the Bonferroni correction threshold.

3.5.2. Individual steroids—In fetal sex-stratified analyses examining individual steroids, we identified several associations with PM_{2.5} during (i) early pregnancy (males: C37183, C48394, C40708; females: C62311, C48394, C37196; Figures S5-S6; Table S4) or (ii) preconception (males: C21294, C47108; females: C37480, C48394). Likewise, a few steroids were associated with temperature during (i) early pregnancy (males: C37198, C37187; females: C37187; Figures S5-S6, Table S4) or (ii) preconception (female: C18474, C37196; Figure S5-S6, Table S4). However, none of these steroids showed sex-differential effects and no associations survived multiple testing comparisons.

3.6. Sensitivity analysis.

We identified statistically significant differences in exposures, maternal age and race/ethnicity between participants included in our main analysis and the remainder of the cohort (Table S1). Main results for both steroid classes and individual steroids remained unchanged after including weights and winsorising steroid levels (Figures S7-S8). In addition, findings for steroid classes and individual steroids were consistent with the main results after the inclusion of pre-pregnancy BMI in the set of covariates (Figures S9-S10).

4. Discussion

In the present study, we leveraged metabolomics data to interrogate associations between PM_{2.5} and a broad array of steroid hormones and their metabolites among an ethnically diverse sample of pregnant individuals living in an urban environment in the Northeastern US. We found that levels of the pregnenolone (early pregnancy exposure only) and androgen (preconception and early pregnancy exposure) steroid classes were positively associated with ambient fine particulate air pollution. We also detected a marginal positive association between early pregnancy exposure and the class of progestin steroids, which remained significant among male pregnancies only. When considering each hormone metabolite individually, we found positive associations between early pregnancy exposure and five steroids, two of which survived multiple comparison testing: 11 β -hydroxyandrosterone glucuronide (a pregnenolone steroid) and androsteroneglucuronide (a progestin steroid).

While a range of indoor air pollutants have been classified as endocrine disrupting chemicals (EDCs) (Darbre, 2018), limited prior research has examined the endocrine disrupting potential of ambient air particulate exposure. Of the studies that have examined hormonal endpoints, the majority have focused on hormones with primary roles in the regulation of the thyroid (Ghassabian et al., 2019; Howe et al., 2018; Zhao et al., 2019) or metabolic systems (Bolton et al., 2012; Sun et al., 2009; Zheng et al., 2013). However, growing evidence has linked PM_{2.5} with a number of adverse reproductive outcomes and disruption of estrogen-related processes (Wang et al., 2021). For example, murine research has shown PM_{2.5} exposure during pregnancy is associated with down regulation of estrogen receptor alpha in uterine tissue via epigenetic mechanisms (Dang et al., 2018), suggesting that PM_{2.5}

might inhibit estrogen synthesis by the ovaries. Aquatic research has also demonstrated toxic constituents bound to PM_{2.5} are associated with dysregulated estrogen and other sex steroids. For example, benzy[a] pyrene exposure has been shown to suppress estradiol and testosterone production and disrupt progesterone levels in the ovaries of female scallops (Tian et al., 2013). In the only human study we identified, non-pregnant female police officers in Italy who had higher exposure to traffic-related air pollution due to traffic-related duties, had plasma estradiol levels that were statistically significant lower compared to a control group of police officers without traffic-related duties (Tomei et al., 2006). While we did not detect any relationship with estrogens in the present study, we did find associations with pregnenolone and androgen hormones. Pregnenolone is the starting material for the production of all steroids, including androgens, estrogens, progestin, and corticosteroids (Little et al., 1971). The first nonreversible step for pregnenolone production is the conversion of cholesterol to pregnenolone. This step occurs in the mitochondria, which has previously been shown to be sensitive to particulate air pollution, of the adrenal cortex, ovary and testis (by the enzyme desmolase) (Brunst et al., 2018; Byun et al., 2016; Colicino et al., 2014; Miller and Auchus, 2011). During pregnancy, pregnenolone is also produced by the placenta, which converts cholesterol to pregnenolone, thus modulating the production of all steroids (Miller and Auchus, 2011). Pregnenolone has two main effects: 1) boosting androgen levels, which can increase metabolism and energy (Craig et al., 2013), and 2) inhibiting cortisol production, a key hormone involved in stress responsivity as well as a range of other functions. For these reasons, pregnenolone is considered an effective stress reducer and clinical application has shown that pregnenolone can protect the brain from acute environmental exposures, such as cannabis intoxication (Vallée et al., 2014). We postulate that a similar buffering mechanism, exerted by the placenta and adrenal gland, may be in place to facilitate the development of the fetus, including the fetal brain, and to protect against the adverse effects of environmental exposures.

Other evidence supporting a putative relationship between air pollutants and reproductive hormone disruption comes from in vitro studies of tobacco smoke exposure, a major source of indoor particulate air pollution (Burguet et al., 2005; Byrne et al., 1991; Hartz et al., 1987; Macdonald et al., 1990; Meikle et al., 1989). Tobacco smoke has been shown to increase levels of several androgens, including androstenedione, and decrease levels of the estrogen estrone, providing a pattern suggestive of smoking-associated inhibition of aromatase activity (Byrne et al., 1991). This is supported by other studies demonstrating that tobacco smoke inhibits the major steroidogenic pathway of granulosa cell aromatase, which plays a key role in the biosynthesis of estrogens, in a dose-dependent manner (Barbieri et al., 1986; Hecht, 2008; Medicine, 2004; Muranaka et al., 1988; Zahran et al., 2007).

Several additional studies have reported on the complex interplay between the induction and inhibition of enzymes involved in steroid biosynthesis following exposure to hazardous components of air pollution, although studies specifically examining PM_{2.5} are sparse. The disruption of this complex enzymatic interplay has downstream consequences for hormone dependent metabolic and biologic processes (Hecht, 2008). For example, polycyclic aromatic hydrocarbons (PAHs) have been shown to induce some microsomal cytochrome *P*-450 enzymes, which metabolize steroid hormones (Barbieri et al., 1986; Kendall et al., 2007) often linked to an anti-estrogenic effect (Michnovicz et al., 1986; Modugno et al., 2003;

Spangler, 1999). In vitro research has also demonstrated that exposure of human trophoblast cells to PM_{2.5} causes reduced progesterone secretion, as well as reduced expression of enzymes central to the conversion of pregnenolone to progesterone (Wang et al., 2017). This finding is supported by similar research that has identified steroid hormone biosynthesis-related gene expression changes following PM_{2.5} exposure, including changes that could specifically result in decreased progesterone levels (Kim et al., 2018).

In sum, our findings contribute to expanding evidence suggesting that air pollution may interfere with steroid release, binding, transport, storage, metabolism, or clearance, resulting in altered circulating steroid concentrations (Muranaka et al., 1988). However, we also cannot rule out effects at the steroid receptor level that alter circulating androgen levels via feedback loops. Additional studies are needed to further clarify the endocrine disruptive role of ambient air pollution, especially during the critical endocrine-dependent pre-conception and early gestational periods. Furthermore, while we did not detect strong associations with ambient temperature, we support future studies that investigate associations between temperature and steroids, including those investigating a wider range of temperature extremes and different parametrizations such as maximum temperature levels. Although prior literature and our study suggested that suboptimal average exposures during preconception and early pregnancy periods can affect maternal endocrine function, further studies should also consider assessing critical windows of exposures by using daily exposure levels.

This study has several strengths. Our exposure models allowed us to estimate levels for both air pollution and temperature at the residential address of each participant. Also, the prospective design ensured temporality and enabled the evaluation of associations of maternal steroids with ambient air particle and temperature exposures occurring during the critical pre-conception and early gestational periods. Additionally, simultaneous examination of different periods of exposure provides a measure against confounding bias, thereby greatly increasing the likelihood that the identified statistically significant associations may be independent of other exposure periods and be causal (Oulhote et al., 2016). Maternal serum was collected at different gestational weeks for different participants and, as expected, steroid levels showed inter-individual temporal trends across pregnancy. While an interesting study design would be to look at the effects of exposure on within-person change in hormone levels across pregnancy, in this study we were limited to measuring metabolite levels at a single time point during gestation. Thus, to account for the expected and observed shifts in hormone levels across gestation, we standardized levels of steroids by week. The use of mixed effect modeling also reduced multiple comparisons testing, which is essential for an unbiased estimation of the associations (Carroll et al., 2006). Although the racial/ethnic diversity of the sample is a strength of this study, it may reduce the generalizability of our findings to more racially homogenous populations. Future studies with larger sample size should investigate the associations within racial/ethnic groups and identify racial disparities during pregnancy and their consequences for newborns, especially given the persistent racial/ethnic disparities in birth outcomes in the US and globally. While we did not demonstrate sex-specific effects for these associations, this may reflect lack of power and this should be further explored in larger samples. Future studies are also needed to evaluate the impact of synthetic steroid drugs during pregnancy on steroid-exposure

associations. While models were adjusted for a number of potential confounders, including several indicators of social and economic status, the possibility of residual confounding cannot be ruled out.

5. Conclusions

Findings from our study suggest that early prenatal PM_{2.5} exposure is associated with higher maternal pregnenolone and androgen concentrations during late pregnancy. In addition, preconception PM_{2.5} exposure may have an increasing association with androgen levels. No sex-difference associations were detected in our ethnically diverse sample. The exposure-steroid association in pregnant women is an important health issue given the ubiquitous nature of air pollution exposure. Identifying modifiable environmental influences on hormone metabolism in pregnancy is important as sub-optimal endocrine adaptation to the state of pregnancy can impact maternal health and perinatal outcomes, and have longer-term consequences for child health.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Data Availability.

On Metabolomics Workbench Study Number: ST001827 Data track ID: 2682. (https://www.metabolomicsworkbench.org/data/show_mwtabfile.php?F=nicolofp_20210610_125247_mwtab_analysis_1.txt).

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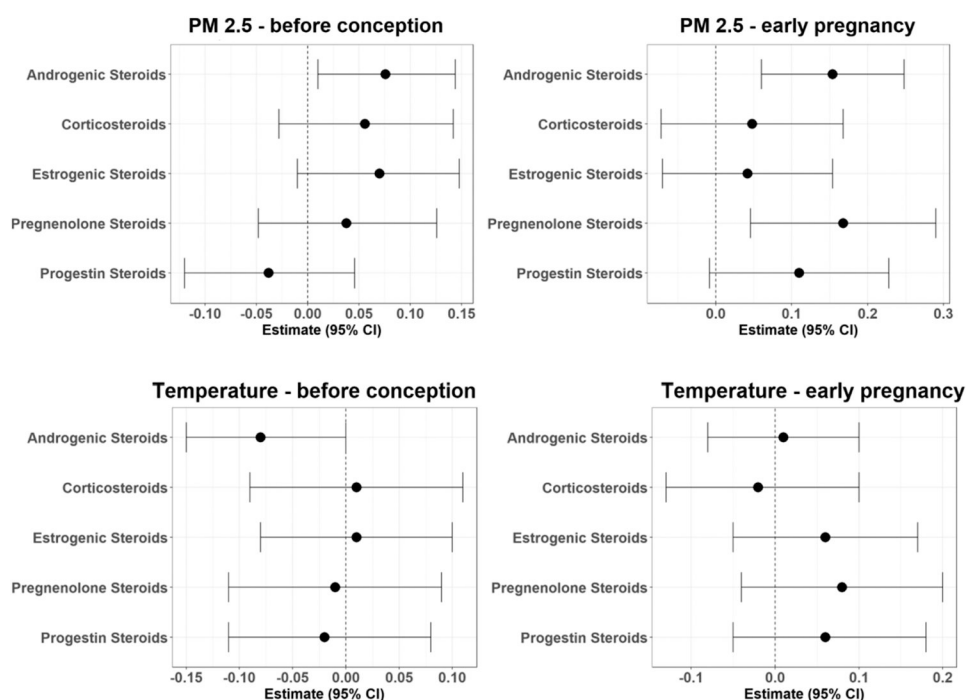
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**Fig. 1.**

Estimates and 95% Confidence Interval (95 %CI) for the associations between classes of steroids and exposure to both PM_{2.5} and temperature during the preconception (three months) and early (20 weeks) prenatal period. * *All robust mixed effect regression analyses included PM and temperature in the same model and were adjusted for maternal age, race/ethnicity, and education; a random intercept for individual was included to account for multiple individual hormones 'repeated' within each class.

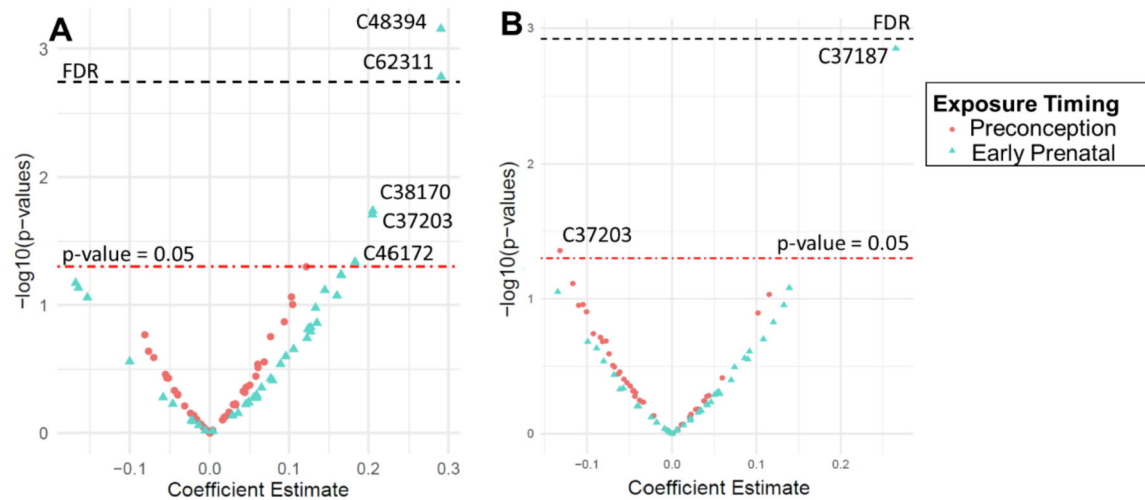
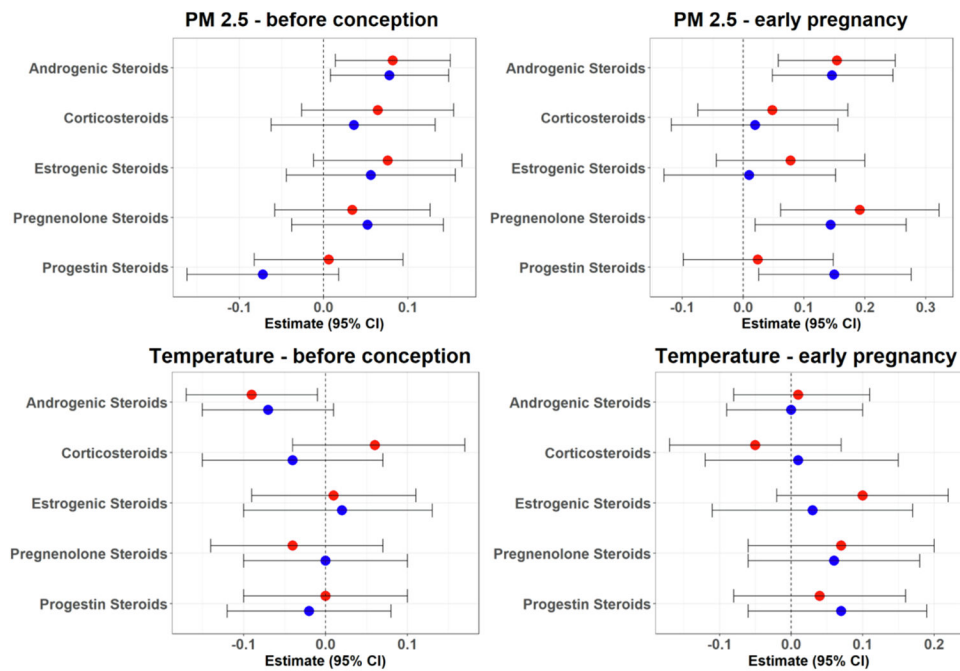


Fig. 2.

Volcano plots showing the robust estimates and $-\log_{10}(\text{pvalue})$ of the associations between individual steroid levels and exposure to **(A)** $\text{PM}_{2.5}$ and **(B)** temperature during early (20 weeks) gestation and (three months) before conception. * *All robust regression analyses were adjusted for maternal age, ethnicity, and education. FDR: False Discovery Rate threshold of 5%. All steroids with $\text{pvalue} < 0.05$ were annotated.

**Fig. 3.**

Sex-stratified analysis. Estimate and 95% Confidence Interval (95 %CI) of the associations between classes of steroids and exposure to PM_{2.5} and temperature during early (20 weeks) gestation and (three months) before conception, stratifying by sex (red: female; blue: male).*

*All robust mixed effect regression analyses included PM and temperature in the same model and were adjusted for maternal age, ethnicity, education levels; a random intercept for individual was included to account for 'repeated' measures within individuals.

Table 1

Participant characteristics – PRISM sample.

Variables	Overall sample (N = 361)	Male* (N = 186)	Female (N = 175)
	Mean ± SD or %	Mean ± SD or %	Mean ± SD or %
PM _{2.5}			
3 months preconception	9.20 ± 1.82	9.19 ± 1.82	9.21 ± 1.83
First 20 weeks gestation	8.89 ± 1.30	8.97 ± 1.33	8.81 ± 1.26
Temperature			
3 months preconception	11.85 ± 8.11	12.08 ± 8.1	11.6 ± 8.14
First 20 weeks gestation	10.43 ± 6.54	10.4 ± 6.67	10.47 ± 6.43
Maternal age (years)	30.77 ± 5.66	30.79 ± 5.55	30.75 ± 5.78
Maternal education			
< 12th grade	20.62%	18.78%	22.54%
≥ 12th grade	79.38%	81.22%	77.46%
Maternal Race/Ethnicity			
White, non-Hispanic/other	37.08%	37.16%	36.99%
Maternal Race/Ethnicity			
Black/Black-Hispanic	29.78%	28.96%	30.64%
Hispanic, non-Black	33.15%	33.88%	32.37%
Fetal Sex			
Female	48.48%	–	100.00%
Male	51.52%	100.00%	–

SD: Standard Deviation; %: percentage.

* Male and female indicate fetal sex.